



ECSA/ECMFA/ECOOOP  
International Workshop on  
Software Engineering for Systems-of-Systems  
Montpellier, France  
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# Issues and Challenges in Ecosystems of Federated Embedded Systems

## EcoFES

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# Agenda



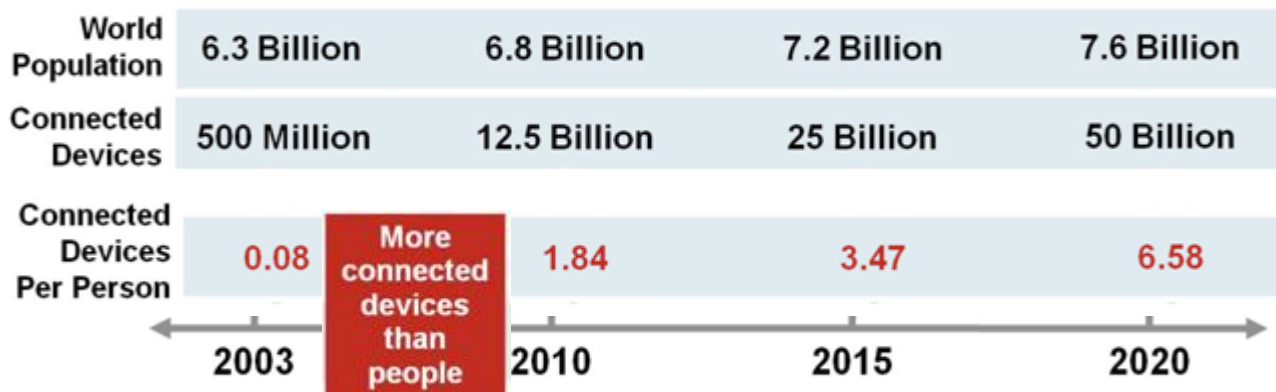
- **Introduction**
- **Federated Embedded Systems**
  - Definition
  - Scenarios of application
- **Research challenges**
- **Current work**
  - Explorative Case Study
  - Preliminary findings
- **Conclusions**
- **Future work**

# Introduction (1/4)

## A view on connected Devices and People



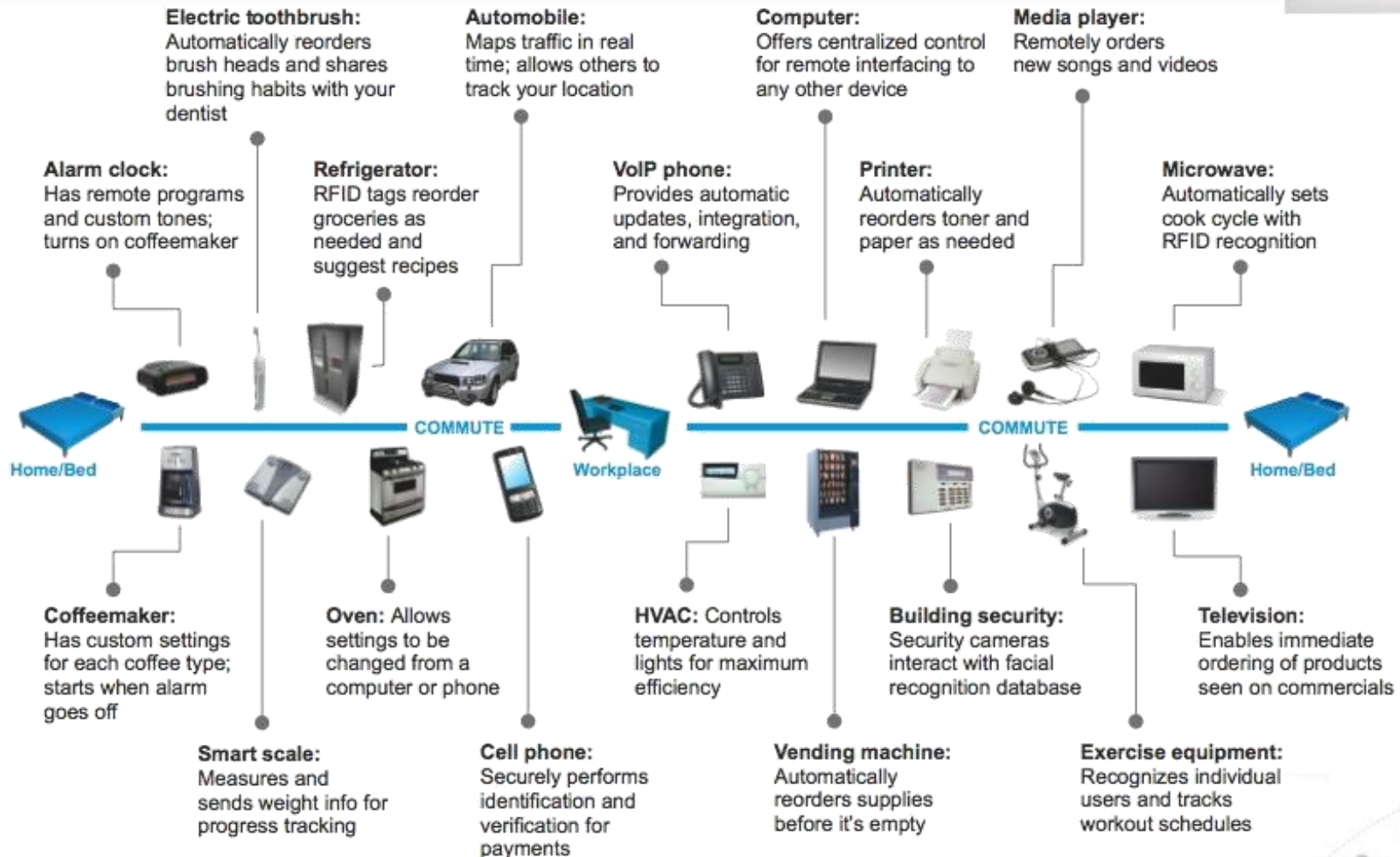
**Our  
World**



Source: Cisco IBSG, April 2011

# Introduction (2/4)

## Connected Devices for Embedded Computing



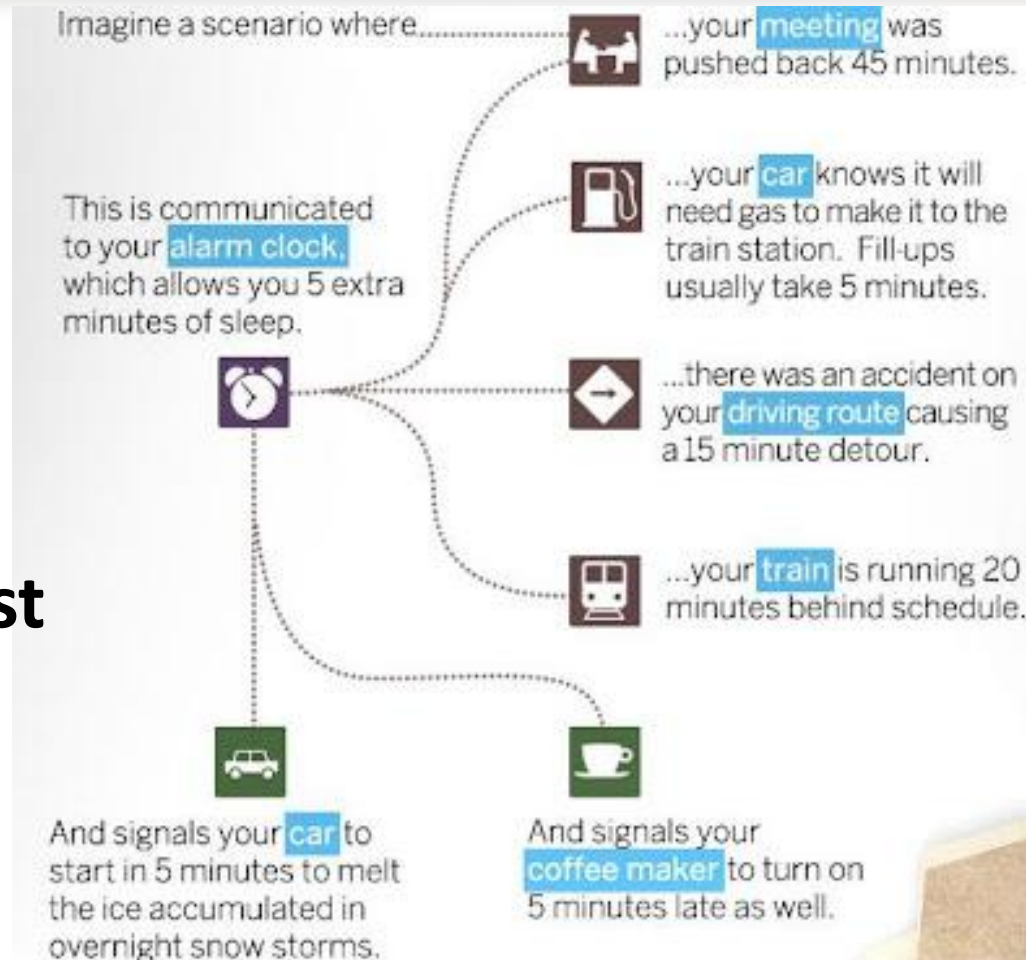
# Introduction (3/4)

## A possible scenario in improving everyday living



### Motivation

..could **adjust**  
and  
**improve**  
our  
schedule



A series of  
**events**  
occurring...

<http://blogs.cisco.com/news/the-internet-of-things-infographic/>



# Introduction (4/4)

## Embedded Systems are Key in today's IoT



- World of “All-Connected-Things”, or Internet-of-Things (IoT)
  - **Systems of Systems (SoS) development:**
    - Combine dedicated complex systems for a specific target
    - Challenge by both industry and academia
  - **Software ecosystems development:**
    - Provide a complementary organizational view to SoS
- **Embedded systems (ES)** → a special case
  - Key components in many everyday systems, e.g.,
    - Automotive
    - Home automation
    - Energy
    - Transportation
    - Healthcare
    - Manufacturing

# Definition of FES <sub>(1/2)</sub>



- **Federated Embedded Systems (FES)**



- A **constellation** of devices that are:
  - **Part of different products**
  - **Control different products**
  - **Exchange data with:**
    - ❖ Each other & various products
    - ❖ External servers

The exchange is made in such a way that ***no individual device is in control over the rest.***

# Definition of FES (2/2)



- **Federated Embedded Systems (FES)**

- Implicitly it means that **different manufacturers** are involved, using different platforms, standards, etc.
- **Structure is dynamic**
  - Deployed, redeployed and extended over time
- A **system** can be **part of several systems** at **different times** or **simultaneously (SoS)**



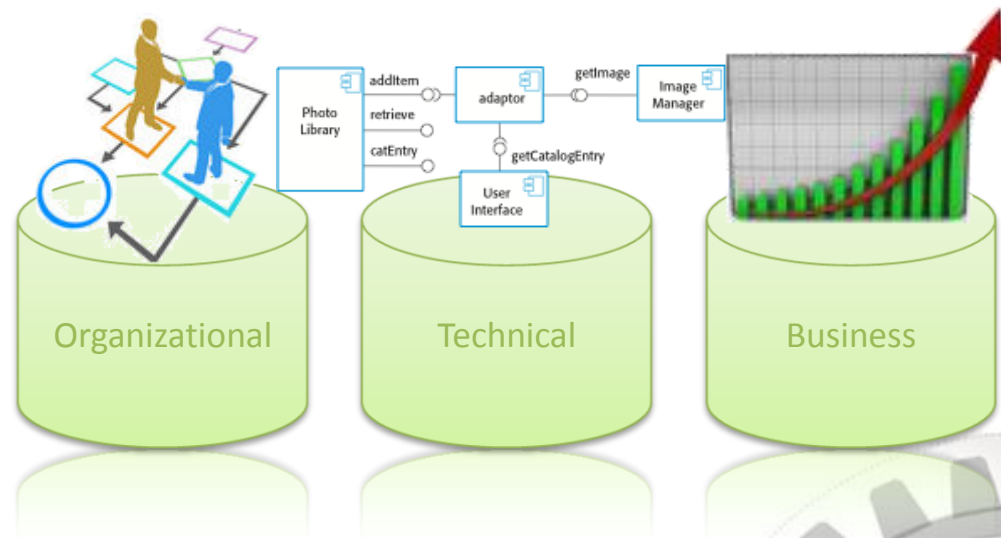


# Federated Embedded Systems

## Various viewpoints

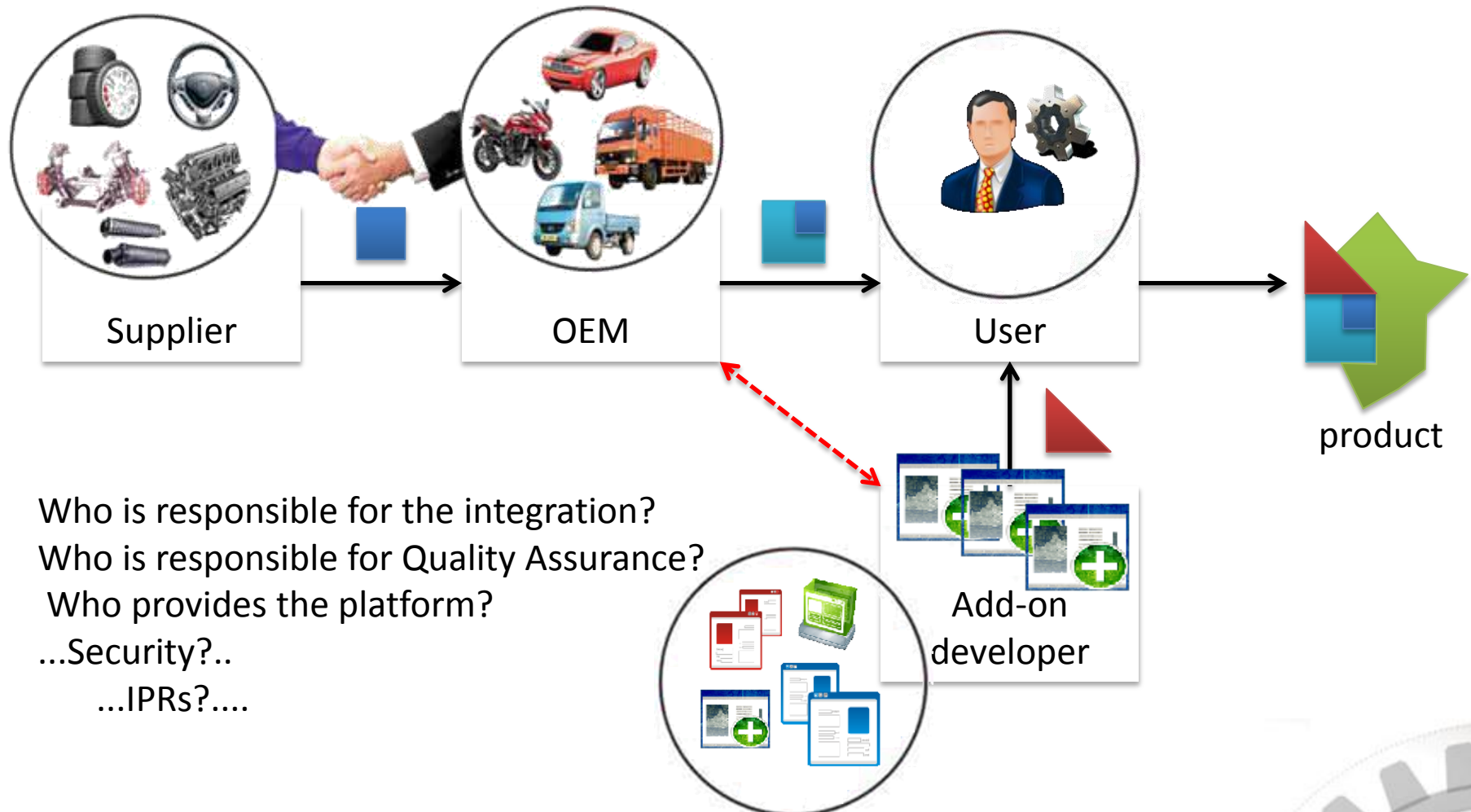


- Several **novel challenges** appear that motivate special consideration
  - **Organizational level**
  - **Technical level**
  - **Business level**



# Scenario in FES development

An example in the automotive industry



Who is responsible for the integration?  
Who is responsible for Quality Assurance?  
Who provides the platform?  
...Security?..  
...IPRs?....

# Topics of research interest



- 1) How would an **innovative open SoS** based on **FES** be **described** and **modelled**?
- 2) How could the **SoS** be **controlled**?
- 3) How would the **virtual players** and **development teams** be **organised**?
- 4) What are the **relationships** between the different actors?
- 5) How would you ensure **quality**?
- 6) How should **technology** be **structured**?
- 7) How are **intellectual property rights** handled?
- 8) What are the potential **business models** that should be established to support the process?

# Research method

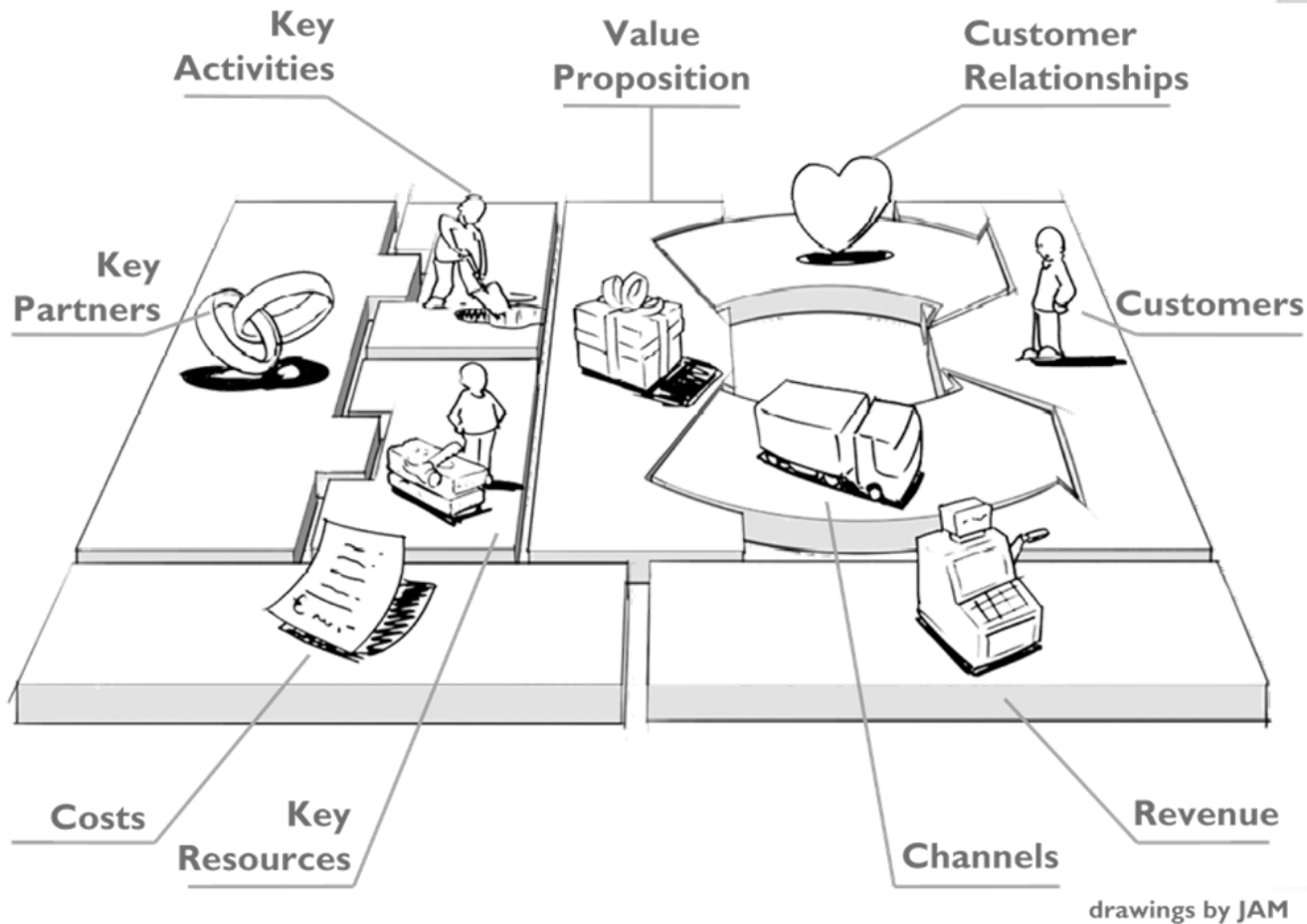


- **Explorative case study** <sup>[1]</sup> (February - June 2013)
  - Interview of **15 senior staff members** at **9 companies**
    - Come from several **industrial domains**
      - **Automotive**
      - **Off-road machinery**
      - **Automation**
    - **Perform and support development and integration** of technical products and services
    - Relate to **different roles** in a future FES
  - **Analysis** → is based on the **Business Model Canvas (BMC)** <sup>[2]</sup>

[1] .Yin. Case Study Research: Design and Methods, fifth ed., SAGE Publications, 2013.

[2] Osterwalder, Pigneur, Business Model Generation, John Wiley, 2010.

# Business Model Canvas [1]



[1] Osterwalder, Pigneur, Business Model Generation, John Wiley, 2010.

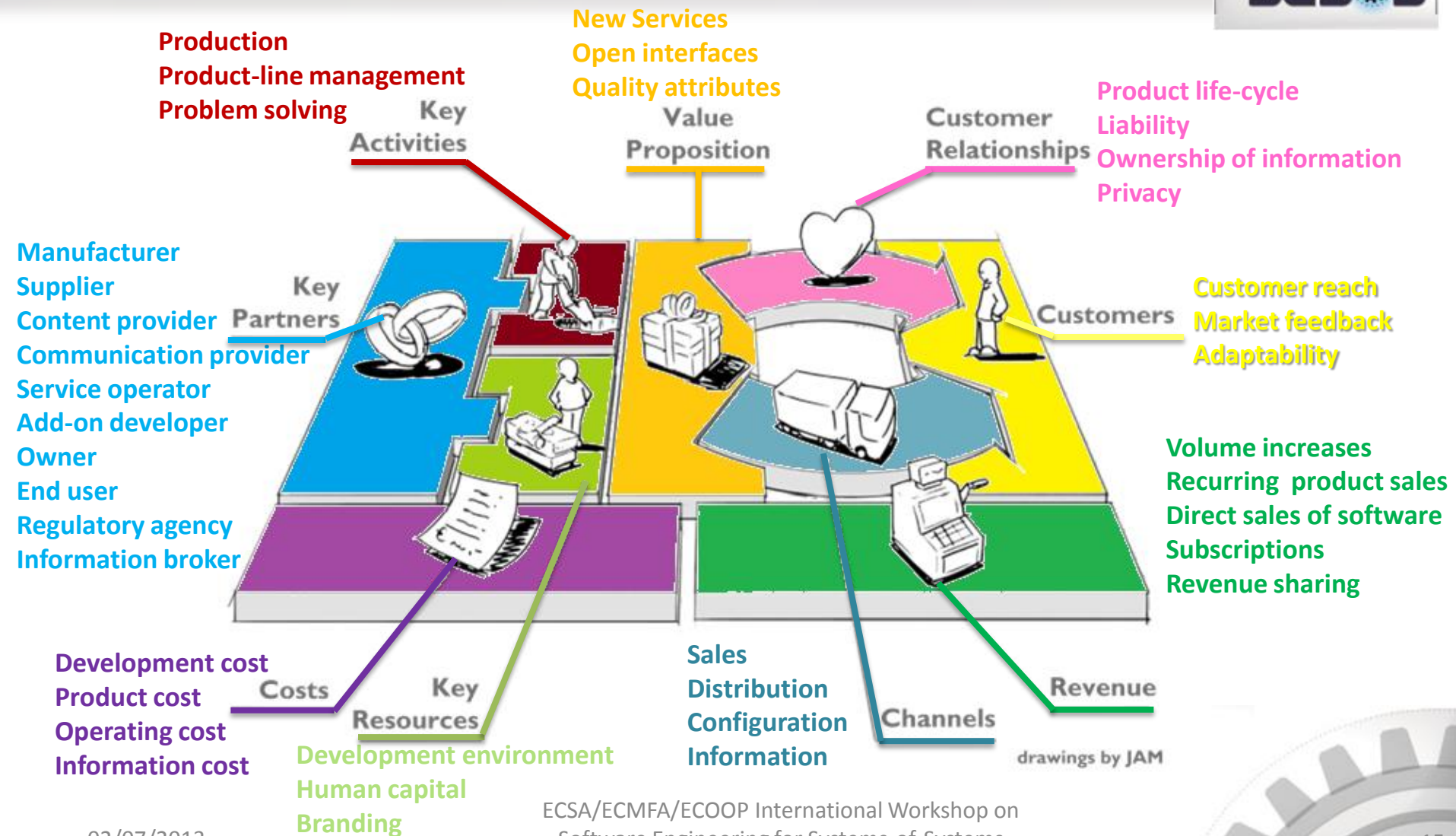


# Preliminary findings (1/2)



| Organizational Level                                                                                                                                                                                                                                                                     | Technical Level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Business Level                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Actors</li> <li>• Roles</li> <li>• Protocols of interaction</li> <li>• Partnerships</li> <li>• Business structure</li> <li>• Tool &amp; platform support</li> <li>• Control</li> <li>• Conflicting interests</li> <li>• Transparency</li> </ul> | <ul style="list-style-type: none"> <li>• Dependability</li> <li>• Distributed functionality</li> <li>• Specifications automation</li> <li>• Interoperability</li> <li>• Data management</li> <li>• Real-time properties</li> <li>• Resource constraints</li> <li>• Asset management</li> <li>• Architecture                             <ul style="list-style-type: none"> <li>○ Adaptable</li> <li>○ Flexible</li> <li>○ Robust</li> <li>○ Reliable</li> <li>○ Reusable</li> <li>○ Complex</li> <li>○ Secure</li> <li>○ Trusted</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>• Competitive landscape</li> <li>• Adoption barriers</li> <li>• Trust</li> <li>• Models formalization</li> <li>• Sustainability</li> <li>• Efficiency</li> <li>• Productivity</li> <li>• Quality</li> <li>• Costs</li> <li>• Strategies</li> <li>• Conflicting targets</li> <li>• Risks</li> <li>• Sharing</li> <li>• Decision making</li> </ul> |

# Preliminary findings (2/2)



# Conclusions (1/2)



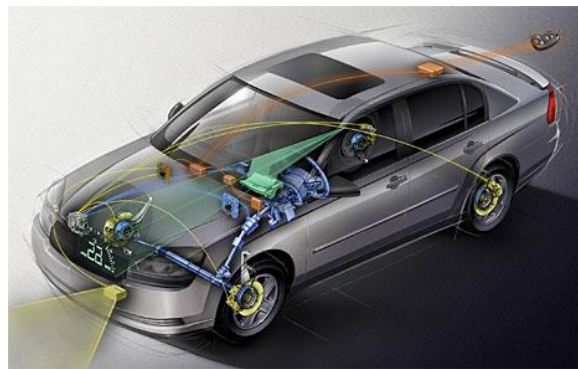
- In support of an **Ecosystem of FES**
  - Proposing a **technical platform** is not sufficient
  - **Business aspects** are equally important, and there is need to identify
    - Main **stakeholders**
    - Implications on **architectures**
    - Implications on **processes, methods, and tools**



# Conclusions (2/2)



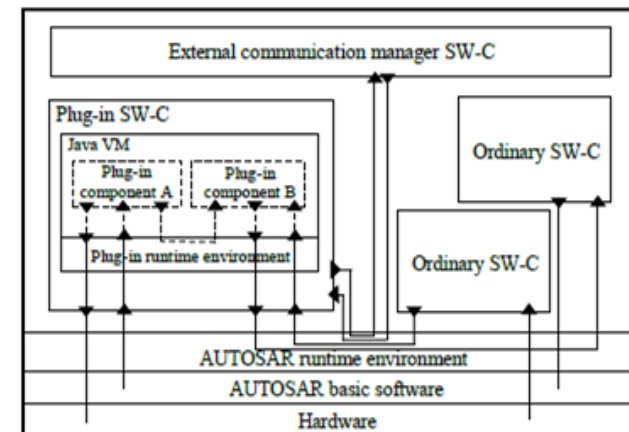
- The **analysis** carried out from interviews made in the **industrial** domain, based on the **BMC**, provides some **answers** to many of the **RQ** relating to **Ecosystems for FES** and identifies:
  - **Main characteristics** of a FES ecosystem
  - **Challenges** for **future** research and practice



# Future work



- Formal **conceptualisation** of the **SoS** for **EcoFES**
  - Prototyping
  - Deployment
  - Commercialization
- ✓ **Designed and implemented** a **dynamic component model** for **reconfigurable AUTOSAR System** [1]



[1] Axelsson & Kobetski, On the Conceptual Design of a Dynamic Component Model for Reconfigurable AUTOSAR Systems, in: Proceedings of the 5th Workshop on Adaptive and Reconfigurable Embedded Systems (APRES'13), Philadelphia, USA, April 8–13, 2013.





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# Any questions?

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# Design of AUTOSAR reconfigurable component model



Application  
Layer

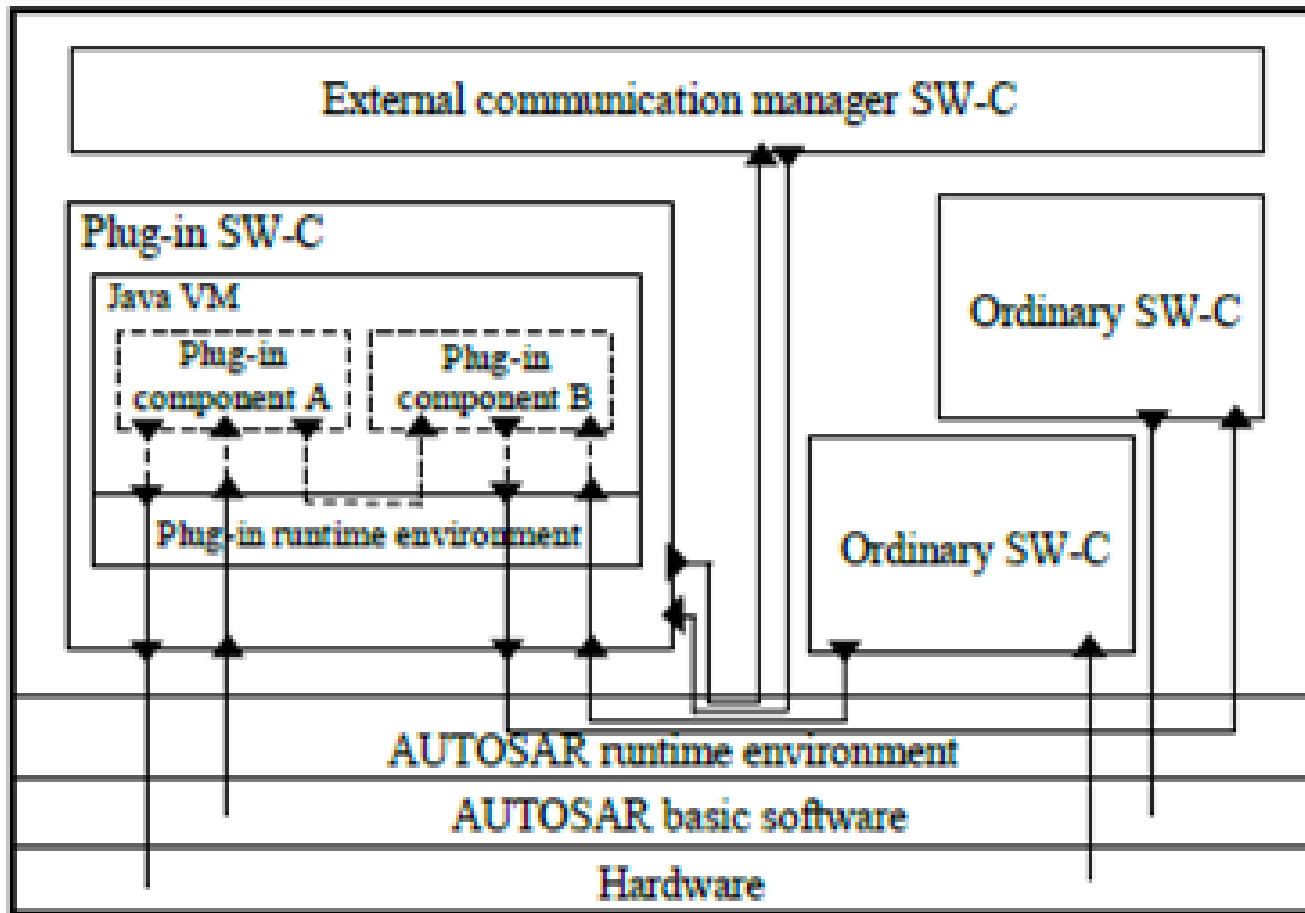


Figure 1. The structure of a dynamic software component model.